

Asymptotic One-Bit Logarithmic Gap

Draft Research Note
for the Conditional Model Version 16.4

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Abstract

This note investigates the asymptotic behavior of the logarithmic gap

$$\Delta_n = \frac{1}{n} \ln \left(\frac{2^{n-1}}{n} \right),$$

arising from the comparison of two real normalizations of a single odd binomial kernel in the conditional model version 16.4.

It is strictly proven that as $n \rightarrow \infty$, the value Δ_n tends to $\ln 2$. After transitioning to the binary logarithm, the corresponding normalized gap tends to 1. This value can be measured as one binary unit, or one bit.

At the same time, it is particularly emphasized that the one-bit limit in itself does not yet turn the logarithmic gap into Shannon entropy. For a strict entropy interpretation, it is necessary to additionally define a probability space and a binary distribution. Therefore, the result should be understood primarily as a precise asymptotic characteristic of the binary structure of binomial coefficients.

Keywords: logarithmic gap; one-bit asymptotic limit; binary logarithmic gap; zero logarithmic gap; odd binomial kernel; binomial coefficients; binary symmetry; two normalizations of a single kernel; residual factors; asymptotic analysis; monotonic sequence; limit of $\ln 2$; binary unit of information; bit; nat; Shannon entropy; information-theoretic interpretation; probabilistic model; coefficient imbalance; quadratic case; conditional model of Fermat's Last Theorem; physical mathematics; philosophy of mathematics.

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1 Initial Definition of the Logarithmic Gap

Let n be a positive integer, and $l > 0$ and $q > 0$ be two positive real residual factors arising from two normalizations of a single odd binomial kernel:

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n.$$

Comparing these two representations yields the exact equality

$$l^n = q^n \frac{2^{n-1}}{n}. \quad (1.1)$$

Since $l > 0$ and $q > 0$, we can define the logarithmic gap

$$\Delta_n := \ln \frac{l}{q}. \quad (1.2)$$

From (1.1) it follows that

$$\Delta_n = \frac{1}{n} \ln \left(\frac{2^{n-1}}{n} \right). \quad (1.3)$$

This equality is a strict algebraic-logarithmic formula. No physical or informational assumptions are used to obtain it.

2 Exact Transformation of the Formula

Using the properties of the logarithm, we transform (1.3):

$$\begin{aligned} \Delta_n &= \frac{1}{n} (\ln 2^{n-1} - \ln n) \\ &= \frac{(n-1) \ln 2 - \ln n}{n} \\ &= \ln 2 \left(1 - \frac{1}{n} \right) - \frac{\ln n}{n}. \end{aligned} \quad (2.1)$$

Equivalently,

$$\Delta_n = \ln 2 - \frac{\ln(2n)}{n}. \quad (2.2)$$

The latter form is particularly convenient for investigating the limit.

3 Limit in the Natural Logarithmic Scale

Theorem 3.1 (Asymptotic limit of the logarithmic gap). *For the sequence*

$$\Delta_n = \frac{1}{n} \ln \left(\frac{2^{n-1}}{n} \right)$$

the following holds:

$$\lim_{n \rightarrow \infty} \Delta_n = \ln 2. \quad (3.1)$$

Proof. From (2.2) we have

$$\Delta_n = \ln 2 - \frac{\ln(2n)}{n}.$$

The standard limit

$$\lim_{n \rightarrow \infty} \frac{\ln(2n)}{n} = 0$$

can be verified by considering the corresponding function of a real variable:

$$\lim_{x \rightarrow \infty} \frac{\ln(2x)}{x}.$$

By L'Hôpital's rule

$$\lim_{x \rightarrow \infty} \frac{\ln(2x)}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0.$$

Consequently,

$$\lim_{n \rightarrow \infty} \Delta_n = \ln 2 - 0 = \ln 2.$$

□

Numerically

$$\ln 2 \approx 0.69314718056.$$

If the natural logarithm is used as a measure of logarithmic information, this value is expressed in *nats*.

4 Monotonic Increase After the Quadratic Case

The limit itself does not show how the sequence approaches it. To clarify this, consider the real continuation

$$\Delta(x) = \ln 2 - \frac{\ln(2x)}{x}, \quad x \geq 2.$$

Its derivative is

$$\Delta'(x) = \frac{\ln(2x) - 1}{x^2}. \quad (4.1)$$

For $x \geq 2$, the following holds:

$$\ln(2x) \geq \ln 4 > 1.$$

Therefore

$$\Delta'(x) > 0 \quad (x \geq 2).$$

It follows that the sequence Δ_n strictly increases for integers $n \geq 2$:

$$0 = \Delta_2 < \Delta_3 < \Delta_4 < \cdots < \ln 2. \quad (4.2)$$

Thus, after the quadratic case, the logarithmic gap becomes positive and then monotonically increases to the finite limit $\ln 2$.

5 Transition to the Binary Scale

To express the same logarithmic gap in binary units, we introduce

$$\Delta_n^{(2)} := \log_2 \frac{l}{q}. \quad (5.1)$$

Since

$$\log_2 x = \frac{\ln x}{\ln 2},$$

we have

$$\Delta_n^{(2)} = \frac{\Delta_n}{\ln 2}. \quad (5.2)$$

Using (1.3), we obtain

$$\begin{aligned} \Delta_n^{(2)} &= \frac{1}{n} \log_2 \left(\frac{2^{n-1}}{n} \right) \\ &= \frac{n-1-\log_2 n}{n} \\ &= 1 - \frac{1+\log_2 n}{n} \\ &= 1 - \frac{\log_2(2n)}{n}. \end{aligned} \quad (5.3)$$

Corollary 5.1 (One-bit asymptotic limit). *For the binary logarithmic gap, the following holds:*

$$\lim_{n \rightarrow \infty} \Delta_n^{(2)} = 1. \quad (5.4)$$

Proof. From (5.2) and (3.1) it follows that

$$\lim_{n \rightarrow \infty} \Delta_n^{(2)} = \frac{\ln 2}{\ln 2} = 1.$$

□

If the value is measured in the binary logarithmic scale, we can write

$$\lim_{n \rightarrow \infty} \Delta_n^{(2)} = 1 \text{ bit.}$$

At the same time, for any finite positive n

$$\Delta_n^{(2)} = 1 - \frac{\log_2(2n)}{n} < 1.$$

Therefore, one is precisely the limit value:

$$\Delta_n^{(2)} \nearrow 1.$$

6 Why Exactly One Binary Unit Appears

The appearance of the unit is determined by the binary structure of the binomial expansion.

The total sum of binomial coefficients is

$$\sum_{r=0}^n C_n^r = 2^n.$$

The sums of coefficients at even and odd positions coincide:

$$\sum_{\substack{r=0 \\ r \text{ even}}}^n C_n^r = \sum_{\substack{r=0 \\ r \text{ odd}}}^n C_n^r = 2^{n-1}.$$

In the second normalization of the odd binomial kernel, the exact mass extracted is

$$2^{n-1}.$$

The first normalization extracts the linear coefficient

$$n = C_n^1.$$

Therefore, the coefficient ratio is

$$\frac{2^{n-1}}{n}.$$

After taking the binary logarithm and dividing by n , we obtain

$$\frac{1}{n} \log_2 \left(\frac{2^{n-1}}{n} \right) = \frac{n-1}{n} - \frac{\log_2 n}{n}.$$

As $n \rightarrow \infty$

$$\frac{n-1}{n} \rightarrow 1, \quad \frac{\log_2 n}{n} \rightarrow 0.$$

Consequently, the one-bit limit does not arise by chance: it is due to the asymptotic dominance of the binary exponential mass 2^{n-1} over the linear coefficient n .

7 Connection to the Zero Gap Condition

Two statements must be strictly distinguished.

The zero gap condition has the form

$$\Delta_n = 0. \quad (7.1)$$

It is equivalent to

$$\ln \left(\frac{2^{n-1}}{n} \right) = 0,$$

which means

$$\frac{2^{n-1}}{n} = 1.$$

Therefore,

$$\Delta_n = 0 \quad \Longleftrightarrow \quad 2^{n-1} = n. \quad (7.2)$$

For positive integers n , this is possible only when

$$n = 1 \quad \text{or} \quad n = 2.$$

In contrast, the asymptotic statement has the form

$$\Delta_n^{(2)} \rightarrow 1.$$

It does not imply

$$\Delta_n^{(2)} = 0$$

and does not yield the equality

$$2^{n-1} = n.$$

Conversely, for $n > 2$, it holds that

$$2^{n-1} > n,$$

therefore

$$\Delta_n > 0.$$

Key Difference

The equality

$$\Delta_n = 0$$

is the condition for exact agreement between the two normalizations and is equivalent to the equality $2^{n-1} = n$.

The limit

$$\Delta_n^{(2)} \rightarrow 1$$

describes a completely different phenomenon: the asymptotic growth of the normalized binary gap after leaving the quadratic case.

8 Information-Theoretic Interpretation

In Shannon's information theory, the entropy of a discrete distribution $(p_1, \dots, p_k)$ is defined by the formula

$$H = - \sum_{i=1}^k p_i \log_2 p_i.$$

For an equiprobable binary choice

$$p_1 = p_2 = \frac{1}{2}$$

we obtain

$$H = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1 \text{ bit.}$$

Therefore, the limit value

$$\lim_{n \rightarrow \infty} \Delta_n^{(2)} = 1$$

numerically coincides with the entropy of a single equiprobable binary choice.

However, this numerical coincidence does not yet imply that $\Delta_n^{(2)}$ is Shannon entropy.

For a strict entropy interpretation, it is necessary to additionally define:

1. a probability space;
2. two possible binary states;
3. the probabilities of these states;
4. a rule connecting the states with the two normalizations of the kernel;

5. a proof that the resulting entropy coincides with $\Delta_n^{(2)}$.

Until these conditions are met, it is mathematically more correct to use the expression

one-bit asymptotic logarithmic gap

instead of the expression

entropy of an arithmetic system equal to one bit.

Research Hypothesis 8.1 (Future Information Interpretation). *The binary structure of the decomposition of the binomial mass into even and odd parts may admit a probabilistic model in which $\Delta_n^{(2)}$ acquires the meaning of an informational defect between two compositional representations.*

This hypothesis is not used in proving the limit (5.4). The limit is a strict analytical result, regardless of its possible physical or informational interpretation.

9 What is Proven and What is Not Yet Proven

Table 1: Logical status of the obtained results

Strictly proven	Requires additional justification
$\Delta_n = \frac{1}{n} \ln \left(\frac{2^{n-1}}{n} \right)$	The necessity of the condition $\Delta_n = 0$ for any hypothetical solution to Fermat's equation
$\Delta_n > 0$ for all integer $n > 2$	The physical status of the logarithmic gap
$\Delta_n \rightarrow \ln 2$	The identification of the gap with thermodynamic entropy
$\Delta_n^{(2)} \rightarrow 1$	The identification of the gap with Shannon entropy
Monotonic increase of Δ_n for $n \geq 2$	Construction of a probability space that generates exactly this value

10 Conclusion

The logarithmic gap

$$\Delta_n = \frac{1}{n} \ln \left(\frac{2^{n-1}}{n} \right)$$

has an exact and fairly simple asymptotic behavior:

$$\lim_{n \rightarrow \infty} \Delta_n = \ln 2.$$

In the binary logarithmic scale

$$\Delta_n^{(2)} = \frac{1}{n} \log_2 \left(\frac{2^{n-1}}{n} \right)$$

we obtain

$$\lim_{n \rightarrow \infty} \Delta_n^{(2)} = 1.$$

Furthermore,

$$0 = \Delta_2 < \Delta_3 < \Delta_4 < \dots < \ln 2.$$

Consequently, the quadratic case is the last point of zero gap, and for all higher exponents, the gap becomes positive and monotonically approaches a finite binary limit.

The value of 1 in the binary scale is naturally connected with the binary nature of the decomposition of binomial coefficients into even and odd parts. It numerically coincides with the entropy of a single equiprobable binary choice.

However, without an independently specified probabilistic model, this result should not be called a proven one-bit entropy, but rather a one-bit asymptotic logarithmic gap.

Finally, this asymptotic result does not eliminate the conditional status of the model version 16.4. To establish a connection with the unconditional proof of Fermat's Last Theorem, it is still necessary to independently prove that any hypothetical positive integer solution must satisfy the condition

$$\Delta_n = 0.$$

Therefore, the most cautious decision is to keep this material for now as a separate research note, without automatically including it in the main version 16.4.